The maximum possible value of g is achieved by selecting $k = 4$ distinct positive integers a_1, a_2, a_3, a_4 of the form $a_i = i \cdot g - 1$ for $i = 1, 2, 3, 4$. The sum of these integers is $10g - 4$, which must be strictly less than $3 \cdot 10^6$. This inequality implies $g < 300000.4$, so the largest integer g satisfying this is $g = 300000$.

For $g = 300000$, the integers are 299999, 599999, 899999, and 1199999, with sum $299999 + 599999 + 899999 + 1199999 = 2999996 < 3000000$. Each term $\frac{P}{a_i} + 1$ (where P is the product of the four integers) is divisible by g because each $a_i \equiv -1 \pmod{g}$, leading to $\frac{P}{a_i} \equiv (-1)^3 \equiv -1 \pmod{g}$, hence $\frac{P}{a_i} + 1 \equiv 0 \pmod{g}$. The greatest common divisor of these terms is exactly g, as verified through modular analysis and examples with smaller g. Thus, the maximum g is $\boxed{300000}$.